SCAFFOLD: Self-Improving Web Agents via Recursive Parametric Skill Abstraction

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Paper version: arXiv 2609.05511v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Web agents learn each task in isolation and discard the procedural knowledge they gather.*

Skill libraries give an **executable** reuse unit, but stay prompt-side.

Motivation: three open gaps

1. **Flat libraries.** Skills cannot call skills: depth is one.
2. **Unbounded growth.** Near-duplicates, no compression.
3. **Prompt-only.** Skills never reach the weights.

Evaluation setup

WebArena · 812 | VisualWebArena · 910 | Online-Mind2Web · 300 | Qwen2.5-VL-7B base | GPT-4o inducer | K = 5 iterations | 30-step budget | 3 seeds

- **All online**; no offline step accuracy.
- **Held out:** Online-Mind2Web split into 9 training domains and 3 test domains.

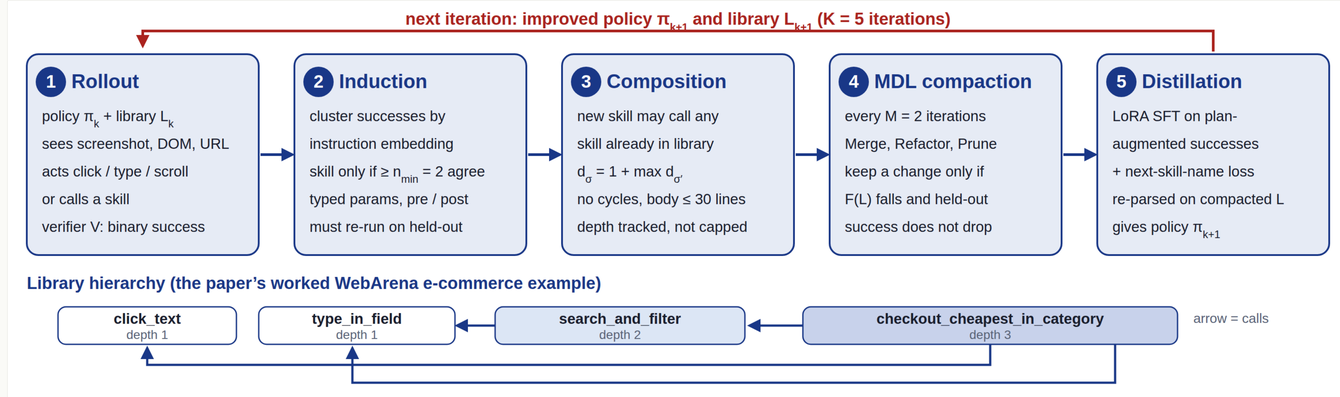
Our read (CUA Radar)

86	★★★	transfer
score of 100	band, ≥ 70	evidence level

- **Reports:** online eval; held-out sites; 3 seeds; matched baselines; cost; ablation; judge check; code.
- **Flags:** needs a ground-truth verifier and a GPT-4o inducer.
- Pattern C19. Authors' numbers, not rerun.

Method: a closed skill loop, rolled out for K = 5 iterations

- **Sees / acts:** screenshot + DOM + URL in; click, type, scroll, wait or skill call out.
- **Skill:** parametric program: typed parameters, pre/postcondition.
- **Signal:** binary verifier success, filtered twice, then distilled into the policy.



Our redraw (not the paper's figure): the five-stage loop and a worked skill hierarchy from the appendix.

$$F(L) = \sum_{\sigma \in L} |\sigma| + \sum_{\zeta \in Z^+} \min_{\text{parse}(\zeta|L)} |\text{parse}(\zeta \mid L)|$$

MDL: library cost + cost of re-expressing successes

Key results: online success rate (%)

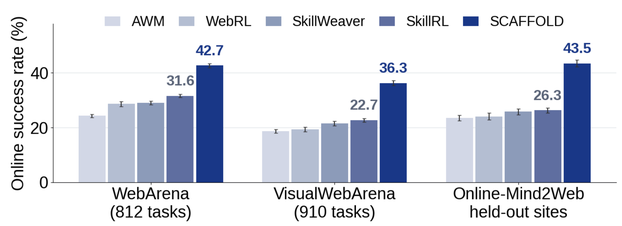


Table 1, redrawn: mean ± std, 3 seeds, Qwen2.5-VL-7B base.

+11.1 / +13.6 / +17.2 pts over SkillRL ($p < 0.05$)

Ablation: WebArena success rate (%)

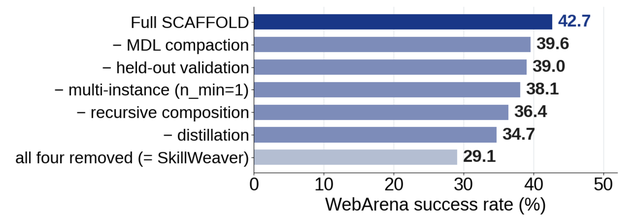


Table 2, redrawn: one component removed at a time.

SO WHAT → *Distillation (−8.0) and recursion (−6.3) cost most; drops sum to 25.7, above the 13.6 gap: synergy.*

Headline Numbers

42.7%
WebArena success rate
vs 31.6% for SkillRL, same base

43.5% held-out OM2W-X
36.3% VisualWebArena
+11.1 pts vs SkillRL

The paper's own controls

CONTROL (WEBARENA)	SCAFFOLD	COMPARED WITH
Step cap 10 / 20 / 30	19.8 / 36.9 / 42.7	SkillRL 13.2 / 26.4 / 31.6
Policy calls / success	5.3	SkillWeaver 9.6
Drop: latency / DOM shuffle	−1.8 / −2.1	SkillRL −3.6 / −4.0
Model judge, not verifier	40.2	42.7 (verifier)

Takeaway

Induce, compose, compact, distill: +11.1 to +17.2 points over the best skill baseline, with a verifier and GPT-4o inducer in the loop.

~470 → 181 skills with MDL | depth 1.05 → 2.71 | reuse 64% | 36 h on 8×A100 per run

Scan to Read

Paper | Code